Flu Trends Analysis

Insights & Recommendations

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INTRODUCTION

Hi everyone,
I'm Yarisel Velacanto, and today I'm going to walk you through my analysis called *Flu Trends Analysis: Insights and Recommendations.*

The main goal here was to take a **data-driven approach** to understand flu mortality trends and vaccination coverage across the country. I wanted to identify where the biggest challenges are and how we can target solutions like staffing and vaccination campaigns to make a real impact.

KEY MESSAGE

By analyzing historical flu trends and vaccination rates, I was able to pinpoint areas that need targeted campaigns and stronger resource planning. This sets the stage for improving public health strategies during flu season.

TRANSITION

First, we'll look at the big picture and see how flu mortality trends have changed over time and what patterns stand out.

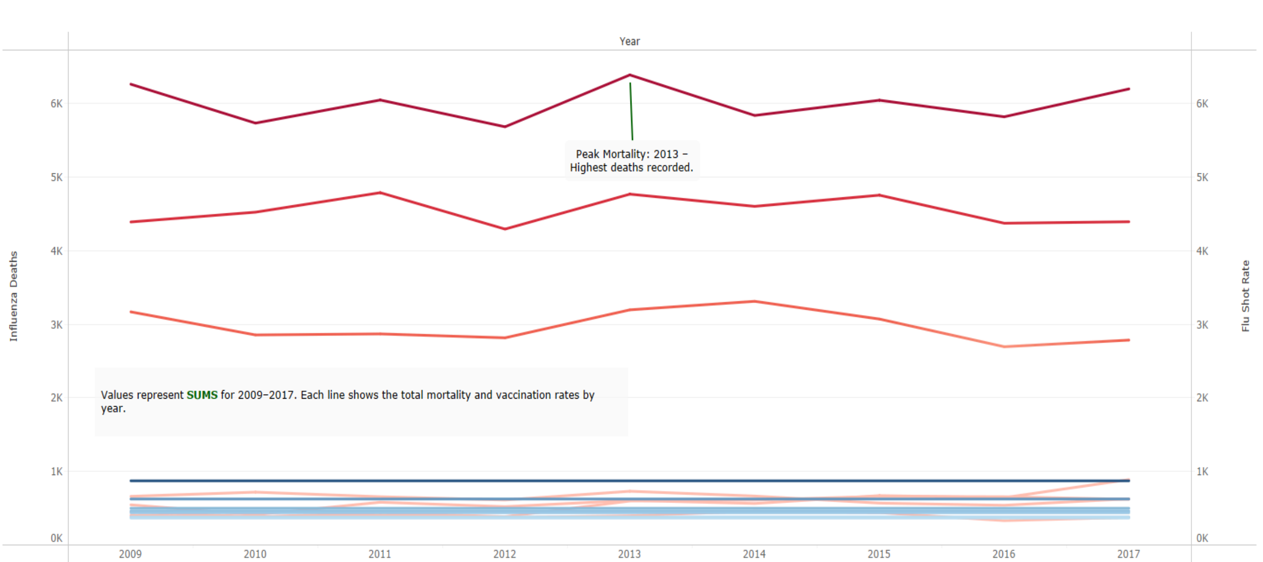

Figure 1. National flu mortality (red) and vaccination rates (blue) from 2009–2017. Data highlights a peak in 2013 mortality and persistent low vaccination rates.

Tracking Influenza Mortality & Vaccination Trends Over Time

Here, I looked at trends in flu mortality and vaccination rates over time, specifically from 2009 to 2017.

ABOUT THE VISUALIZATION

The chart shows two key patterns:

- The red lines represent flu-related deaths, with a visible spike in 2013, the worst year in this period.
- The blue lines represent vaccination rates, which have remained relatively low despite those mortality spikes.

KEY INSIGHTS

Vaccination rates haven't improved enough to reduce flu deaths, especially during critical years like 2013. This shows there's a coverage gap that needs to be addressed through stronger outreach and early-season campaigns.

TRANSITION

Now that we've seen how trends shift over time, let's explore how this looks across different states and see whether vaccination rates truly connect to mortality outcomes.

Flu Vaccination Rates & Mortality Analysis by State

This part of the analysis focuses on how vaccination rates and flu mortality vary across different states.

I used two visuals here:

1. A **scatter plot** showing the relationship between flu shot rates and mortality.
2. A **map** showing vaccination coverage and death rates across the country.

EXPLANATION OF THE VISUALIZATIONS

On the scatter plot, we can see a clear negative trend. States with lower vaccination rates tend to have higher mortality.

The map gives a clearer regional picture:
• Green shading shows vaccination coverage (darker means better coverage).
• Red markers show where flu deaths are highest.

KEY DISCREPANCIES

Some states really stood out:

- Texas and Mississippi: Low vaccination rates and high mortality. These are clear areas that need stronger outreach campaigns.
- Wyoming: High vaccination rates and low mortality. This shows what success can look like.
- Colorado: Decent vaccination coverage but still high mortality, which suggests other factors like healthcare access or reporting accuracy play a role.

KEY MESSAGE

This analysis shows that while vaccination rates are important, they are not the whole story. In states like Colorado, other barriers such as healthcare access and population density may also affect mortality.

TRANSITION

With these insights in mind, let's talk about the key takeaways and what actions we can take to make a real difference.

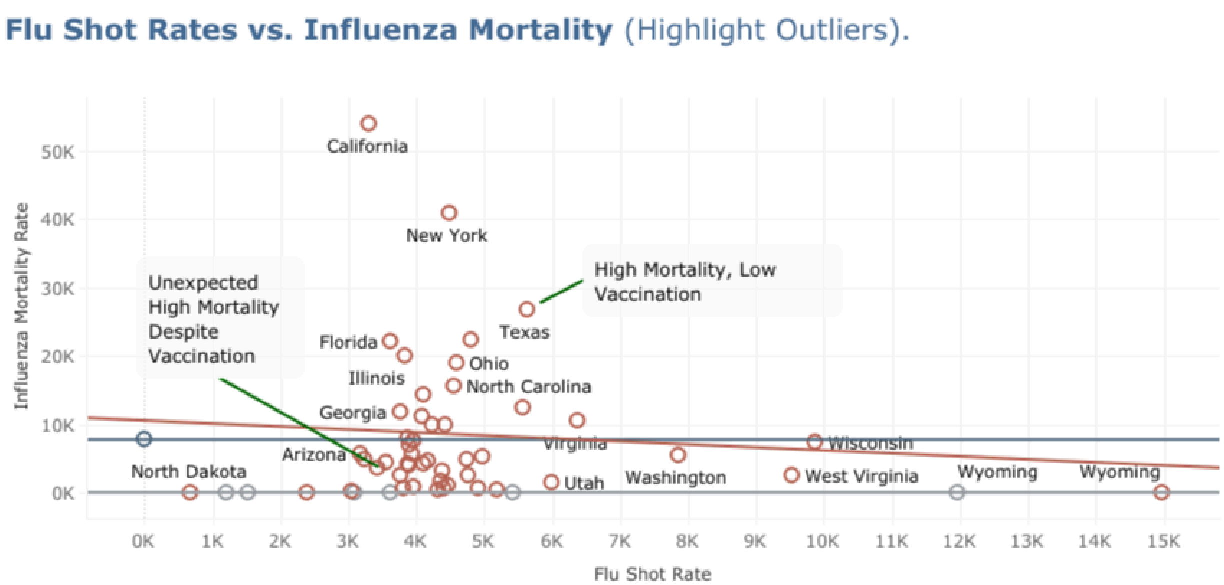

Figure 2A. Relationship between vaccination coverage and influenza mortality by state.

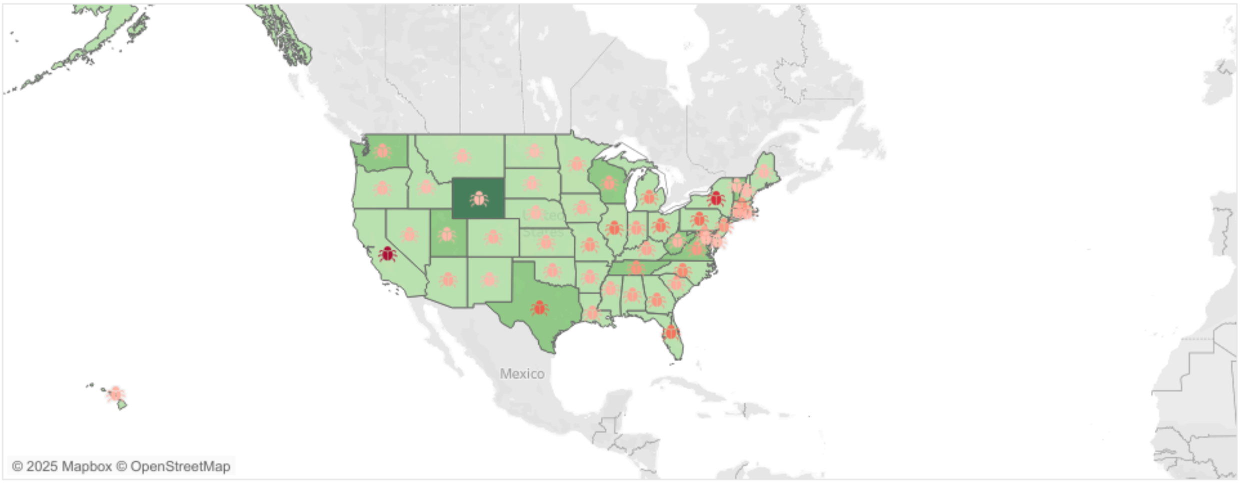

Figure 2B. U.S. map showing vaccination coverage (green shading) and mortality (red markers).

Key Takeaways and Strategic Recommendations

Let’s take a step back and summarize the biggest takeaways from this analysis and what actions we can take to address the challenges we’ve seen.

KEY FINDINGS

- **Low Vaccination Rates** = **Higher Mortality:** States like Texas and Mississippi show a clear pattern of higher deaths linked to lower vaccination coverage.
- **Persistent Mortality Despite Vaccination:** States like Colorado and Washington still face high mortality even though their vaccination rates are fairly strong. This suggests deeper issues like healthcare access, population density, or reporting gaps.
- **Resource Hotspots:** States like California consistently record the highest mortality rates, which means they need extra support and staffing during flu season.

RECOMMENDATIONS

Here’s what I suggest based on these findings:

- **Prioritize Vaccination Campaigns:** Focus on states with low coverage such as Texas and Mississippi. Launching campaigns before peak flu months to improve vaccination rates and save lives.
- **Resource Planning for High-Mortality States:** Increase staffing and supply allocation for high-risk areas like California during peak months.
- **Investigate Other Barriers:** For states like Colorado, conduct deeper research into healthcare access, population density, and reporting gaps to understand why

mortality remains high.

KEY MESSAGE

By focusing resources where they are needed most, we can improve vaccination coverage, strengthen prevention strategies, and save more lives during flu season.

TRANSITION

Next, I will share the forecast chart that shows how vaccination and mortality trends are expected to change, along with how this can guide planning for upcoming flu seasons.

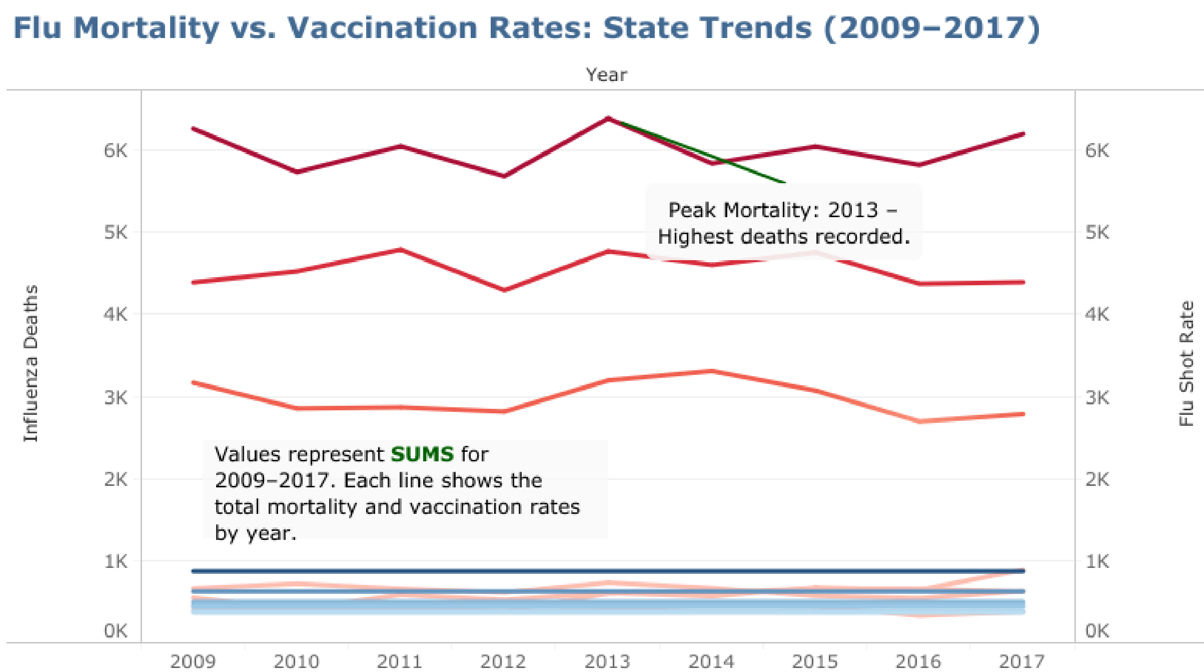

Figure 3A. Trends in flu mortality and vaccination rates from 2009 to 2017. This chart highlights that states with consistently lower vaccination rates also experience higher mortality, with 2013 marking the most severe flu season within this range.

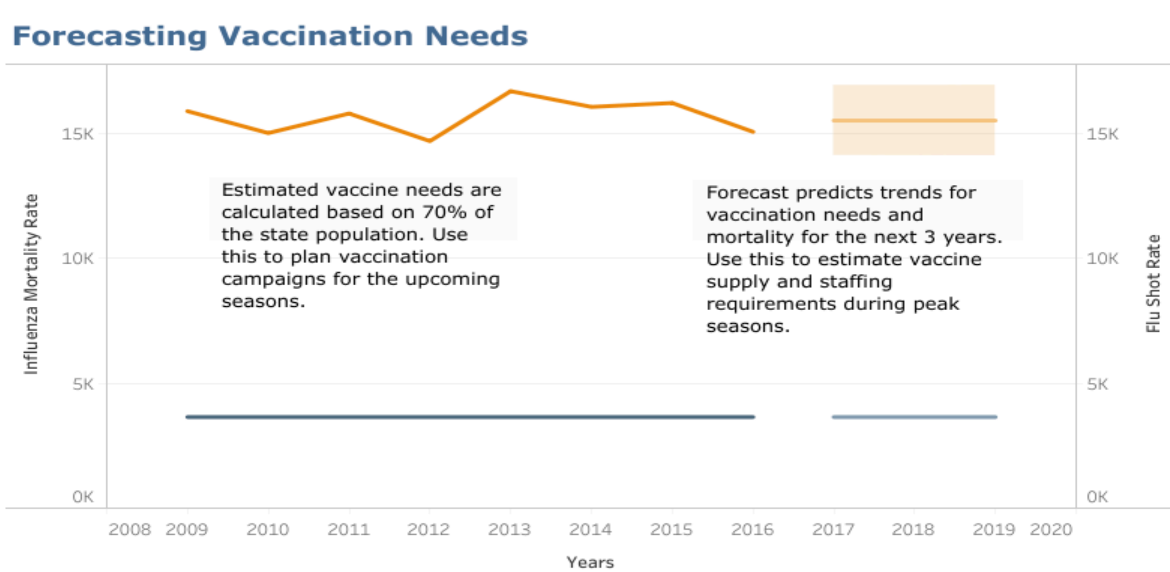

Figure 3B. Forecasted vaccination needs and mortality rates for upcoming flu seasons. The projection estimates future vaccine demand and staffing requirements based on current patterns, helping public health teams prepare for peak seasons and allocate resources effectively.

Reflection on Flu Trends Analysis and Recommendations

DATA LIMITATIONS

While this project revealed valuable insights, there were a few limitations worth noting:

- **Missing Data:** Some states had incomplete data for vaccination rates or flu deaths, which could affect accuracy.
- **Lack of Demographic Information:** Without details such as age or income, it was not possible to identify which populations were most at risk. This information could have helped target recommendations more effectively.
- **Seasonality:** The dataset did not include specific seasonal breakdowns. Having month-by-month data would have helped pinpoint when flu deaths spike, allowing for better staffing and vaccination scheduling.
- **Reporting Bias:** Some states may not report flu deaths or vaccination rates as accurately, which could skew results and make comparisons harder across regions.

METRICS FOR MONITORING IMPACT

To measure the success of these recommendations, here are the key metrics I would track moving forward:

- **Vaccination Rates:** Monitor yearly changes in vaccination coverage, especially in states like Texas and Mississippi.
- **Staffing Levels:** Track staff-to-patient ratios in high-risk states such as California during peak flu months.
- **Hospitalization and Mortality Rates:** Observe whether flu-related deaths and hospitalizations decrease in areas where interventions are implemented.
- **Healthcare Access:** Evaluate improvements in access to flu shots, clinic availability, and reduced wait times.

WHAT THIS MEANS

These limitations highlight the importance of stronger and more consistent data collection across all states.

Even with the gaps, tracking these metrics will help determine whether the recommended strategies are making a real difference.

If more complete and detailed data were available, especially demographics and seasonal patterns, it would open the door to more precise and targeted solutions that could save even more lives.

CLOSING NOTE

By applying data-driven insights to public health decisions, we can turn numbers into action and ensure that every vaccination counts toward a healthier and more resilient population.